Week 2 | Heart Sound Characteristics

Smart Digital Stethoscope | EEE 495 | Module A: Acquisition hardware and characterisation
Prepared by: Kaan | 6 October 2026 | Amplitude range, frequency band and interference

This work is a student project and makes no diagnostic claim. This report is a literature-based study. Project measurements will use an acoustic phantom only; no human recordings or experimental results are presented here. [1,3]

1) Weekly objective and signal being studied

The Week 2 Module A task is to study heart sound characteristics: **amplitude range, frequency band and sources of interference**. These characteristics explain what the acquisition chain must preserve. Microphone model selection belongs to Week 3; numerical gain, bandwidth, noise-floor and clipping-margin requirements are developed in Week 4. [1]

A phonocardiogram (PCG) records mechanical heart sounds. S1 is associated with atrioventricular valve closure near the start of systole; S2 with semilunar valve closure near the start of diastole. Their sequence and waveform must remain distinguishable in the recorded signal. [6]

2) Amplitude: the measurement point and unit matter

Quantity	Meaning and conditions to record
Acoustic pressure	Pa or mPa at the microphone/coupler; state RMS or peak, frequency band, averaging interval, calibration and coupling geometry.
Sound pressure level	L_p = 20 log10(p_RMS / 20 microPa), in dB SPL. State whether the level is unweighted or frequency-weighted. [5]
Electrical amplitude	Microphone/AFE output in volts; state measurement point, bias removal, RMS/peak convention, gain and bandwidth. Sensitivity alone is not the heart-sound amplitude.
Digital amplitude	ADC counts or level relative to full scale (dBFS); state scaling and peak/RMS convention. Uncalibrated audio samples cannot be interpreted directly as Pa.

3) Published amplitude range and its limits

Longhini et al. studied S1 in **20 healthy subjects aged 17-29**, using calibrated equipment with a **6 mL air cavity** at apical and mid-precordial sites. Their mean unweighted spectra reached about **80 dB SPL at 12-20 Hz** and declined to about **35 dB SPL at 110-120 Hz**. B-weighting changed the reported maximum to 40-50 dB at 20-60 Hz. [4]

Thus, approximately **35-80 dB SPL is a reported span across frequency components in that setup**, not a universal range of whole-heartbeat RMS levels, a population minimum/maximum, or our phantom's measured output. The study concerns S1; it does not establish an amplitude range for every heart sound. [4]

Calculated level examples [5]	Equivalent RMS pressure
40 dB SPL	0.002 Pa = 2 mPa
60 dB SPL	0.020 Pa = 20 mPa
80 dB SPL	0.200 Pa = 200 mPa

These are unit conversions, not measurements or a prescribed test range. A 20 dB increase means ten times the pressure amplitude. For a single frequency, V_RMS = S(f) x p_RMS; microphone sensitivity S(f), coupling and AFE gain determine the voltage delivered to the ADC. Our input-voltage range remains to be measured.

Frequency Band and Low-Frequency Preservation

Week 2 | Module A | Relating the signal spectrum to acquisition behaviour

4) Project band and its interpretation

Frequency region	Interpretation for this project
Below 20 Hz	Not in the required characterisation band. Very slow motion/pressure drift can be large. Some cardiac mechanical energy may also occur here; the project boundary is not a claim that all sub-20 Hz content is noise. [1,4,7]
20-150 Hz	Priority region highlighted by the component document. Preserve low-frequency content rather than adopting a speech-oriented response that suppresses it. [2]
150-500 Hz	Still part of the required response measurement; retain and characterise this region rather than reducing the project to the priority band alone. [1]
Above 500 Hz	Outside the mandatory 20-500 Hz characterisation band. This is a project boundary, not a universal physiological upper limit. The wider-band study in [7] also investigates content above 500 Hz.

The 20-150 Hz emphasis and the 20-500 Hz measurement band serve different purposes and are compatible. The former identifies the especially important low-frequency region; the latter defines the full required characterisation range. [1,2]

Heart rate is not acoustic frequency. For example, 60 beats/min corresponds to one cardiac cycle per second. It does not mean the sound waveform has only a 1 Hz frequency component. S1/S2 are short events with oscillatory content distributed over a frequency band.

5) Why a high lower cutoff loses useful information

Every stage can alter the low-frequency response: microphone, acoustic seal/cavity, coupling capacitors, analogue filters, ADC processing and software filters. A leak or changing seal can change the response even when the electronics are unchanged. Published coupler research identifies microphone/electronics and acoustic leakage as determinants of low-frequency behaviour. [7]

For an ideal first-order high-pass stage, $|H(f)| = f / \mathrm{sqrt}(f^2 + f_c^2)$. The following calculated example compares attenuation at **20 Hz** for three possible cutoff frequencies; it is not a measurement of a selected component.

High-pass cutoff f_c	Calculated gain at 20 Hz
10 Hz	-0.97 dB
20 Hz	-3.01 dB
100 Hz	-14.15 dB

A 100 Hz cutoff in this example strongly attenuates the lower end of the project band. Even a 20 Hz cutoff already gives approximately -3 dB at 20 Hz. Cascaded stages add attenuation in decibels, so checking only one component is insufficient. A frequency-range label beginning at 20 Hz does not establish the response tolerance there.

6) Engineering implications of amplitude and bandwidth

Weak components require low in-band noise, while large S1/S2 peaks and contact transients require headroom. Increasing gain raises both the useful signal and upstream interference and may cause clipping. RMS level describes energy over a stated interval; peak level is needed to assess clipping. A heartbeat is not a sine wave, so its peak/RMS ratio must not be assumed to equal sqrt(2).

For later phantom work, record the reference file and playback level, contact conditions, acquisition gain, filter settings and calibration. Preserve unnormalised raw samples. Compare amplitudes only with these conditions known; independently normalising each recording would conceal level differences. Numerical requirements will be set in Week 4. [1,3]

Interference Sources and Study Conclusions

Week 2 | Module A | Mechanisms, observable effects and planned controls

7) Sources of interference

The mechanisms below follow the project's interference concerns and acoustic-coupling evidence. The checks are proposed engineering controls, not completed experiments. Several disturbances overlap the useful band, so filtering alone cannot recover all lost information. [1,2,7]

Source	Effect on the recording	Control / later check
Contact and motion	Rubbing, cable movement or changing pressure produces drift and impulses; large transients may overload the sensor or AFE.	Fix the chestpiece and cables. Compare stationary and deliberately disturbed phantom recordings at the same gain.
Speech / ambient sound	Airborne sound enters through the head, leaks or housing and can overlap the cardiac band.	Use a quiet baseline; inspect seals. Compare controlled background sound on/off with unchanged phantom excitation.
Mains interference	50 Hz pickup and harmonics enter via cables, supply coupling or grounding. The fundamental lies inside the priority band.	Use short shielded analogue wiring, controlled grounding and clean supply routing. Inspect the raw spectrum before a notch filter.
Mechanical coupling changes	Changes in seal, cavity, layer compression or placement alter amplitude and frequency response, even without extra audible noise.	Use an indexed rigid holder; record contact pressure, layer thickness and placement. Repeat removal/reseating checks.
Sensor / electronics noise	Microphone self-noise and AFE/ADC noise can mask weak components; low-frequency noise is relevant.	Measure a quiet acoustic baseline and a separate terminated-input AFE baseline in a stated bandwidth.
Playback / biological background	Phantom amplifier/exciter distortion can be recorded as signal. Published source recordings may already contain breathing or other body sounds.	Retain the reference file and playback settings; check clipping. Compare source and acquired records without collecting new human data.

A 50 Hz notch also removes useful signal around 50 Hz; a high-pass filter may suppress both motion and low-frequency heart sound. Hardware/mechanical prevention and raw-data retention are therefore needed before judging filtered recordings. Coupler research also reports microphone saturation without ADC saturation: absence of clipped digital samples alone does not prove that the entire chain remained linear. [7]

8) Week 2 outcome

The study establishes three points: amplitude must be reported with units and measurement conditions; the full project band is 20-500 Hz with special attention to 20-150 Hz; and contact, ambient, electrical and coupling disturbances can mask or distort that signal. These findings provide the basis for Week 3 sensor comparison and Week 4 requirements. No microphone purchase decision or measured phantom amplitude is claimed in this Week 2 report. [1,2]

References

[1] Advisor, SDP4_Digital_Stethoscope_Plan_EN.docx / SDP4_Digital_Stethoscope_Plan_TR-1.docx: project scope, mandatory measurements and EEE 495 Module A Weeks 2-4.

[2] Advisor, SDP4_Component_Requirements_EN.docx: low-frequency emphasis and interference precautions. [3] SDP_Assessment_Criteria_for_Students.docx: reporting, reproducibility and evidence rules.

[4] C. Longhini et al., "Phonometric Study of the Human First Heart Sound," Japanese Heart Journal, 23(5), 711-716, 1982. doi:10.1536/ihj.23.711. Methods and mean spectra; external literature, not project measurements.

[5] Praat documentation, Sound pressure level: RMS pressure, 20 microPa reference and calibration.

[6] J. Oliveira et al., The CirCor DigiScope Phonocardiogram Dataset, PhysioNet, v1.0.3, 2022, Background. doi:10.13026/tshs-mw03. Used here for S1/S2 context; no dataset measurements performed.

[7] H. Zimmermann et al., Acoustic Coupler for Acquisition of Coronary Artery Murmurs, Computing in Cardiology 38, 209-212, 2011. Coupling, dynamic range and interference evidence.